

Defect Processing and Lifecycle Procedure

Version 9.0

P3402

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Document Control

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Revision History

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1 Apr 05	4.0	Initial version (based on merging of existing QMS)	R. Russell
29 Aug 05	4.1	Updated with details on Bug Review process	H. Obermeyer
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03 Jun 08	4.5	Added Root Cause Analysis to Bug Review Team responsibilities. Added requirements to use T3402 Bug Review Meeting Minutes Template. Added requirement to post meeting minutes in Bug Review Workroom. Added high level summary of Root Cause Resolution Process	R. Russell
28 Jun 2011	4.6	Changed name of procedure to Defect Processing & Lifecycle Defined a quorum needed to hold meeting Defined Project Core Team (PCT) Defined changes to the process and Azure DevOps database	A. Dickinson
17 Dec 2012	5.0	Modified to match new Maintenance Initiative and to be consistent with common defect definitions agreed to with CS&T.	K. Young
Nov 2014	5.1	The following updates have been implemented to the procedure: - included definition for Impact and Severity - clarified guidelines for procedure maintenance, Defect Priority and Root Cause Analysis.	T. Miller
1 Nov 2016	6.0	- Updated Defect Definition - Included the established Defect Triage Guidelines - Clarified requirements for Root Cause Analysis - Provided Defect Escalation guidance - Updated the SLO Guidelines	T. Miller
22 May 2018	7.0	Update to reflect new Azure DevOps Defect Management system and record work item type reflecting the fix of the defect to be delivered in Primary Version in development and the Supplemental release version(s).	T. Miller
Mar2021	7.1	Periodic Update. Added Executive Awareness/Escalation reference. Reference to Security Champion added. Numerous changes to incorporate Azure DevOps process/State & Actions.	A. Dickinson
8 Oct 2021	8.0	Major Update - Removed Definitions to reference the QMS Glossary - Streamlined the procedures to be on the workflow/process	T. Miller

20 Oct 2022	9.0	Major Update - Added the criteria when the customer issue is escalated to the CAPA System. - Added the UX role to the defect process	T. Miller
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1. Introduction

1.1 Purpose

The purpose of the Defect Processing procedure is to:

- Provide a standard process for moving and tracking software defects through their lifecycle.
- Establish roles and responsibilities for various people involved in the lifecycle.

Furthermore, the purpose of the Defect Triage, which is an integral part of Defect Processing, is to:

- Provide a standard process for reviewing, triaging, and tracking software defects and allow defects to be prioritized with balanced view from all groups, including the field view from Customer Support on impact of a defect on customer satisfaction/experience.
- Establish roles and responsibilities for Defect Triage team members
- Identify and prioritize defects for an existing or future release.
- Review and escalate customer enhancement requests submitted via Aha! to the Customer Enhancement Team (CET) for disposition. Customer enhancements will be processed in accordance with the P3004-Customer Enhancement Management Procedure.
- Facilitate on-going communication between Customer Support, Research and Development (R&D), Quality Engineering (QE), and Product Management (PM).
- Allow Customer Support to provide feedback to customers, in a timely manner, regarding the disposition of their reported defect(s)
- Facilitate the management of the defect backlog, to ensure that any limits exceeded are reviewed. Backlog limits may be set Senior Management from Product Management, and Value Management Organization with approval by Executive Vice President of Technology.

1.2 Scope

This procedure applies to all software and documentation defects for products developed by Research & Development, including shipped product versions and product versions under development and applies to all locations within the scope as defined in the Quality Manual.

1.3 Definitions

Refer to the Quality Management System (QMS) Glossary for all QMS related terms, abbreviations, and definitions.

1.4 Related Documents

Document	Document ID#
Quality Manual	QMS_M00
QMS Glossary	QMS_GLO
AspenTech Agile Procedure	P2501
AspenTech PDLC Procedure	P3000
AspenTech Agile Procedure	P3003
Cumulative Patch Procedure	P3701
Emergency Patch Procedure	P3702
Customer Enhancement Management Procedure	P3004
<u>Defect Triage Process Guidelines and Best Practices</u>	N/A
Executive Awareness & Escalation Procedure	P3705
Cyber Security Procedure	P4403

2. Roles and Responsibilities

Defects are managed and controlled in the electronic defect management system of the Application Lifecycle Management system. The roles and responsibilities of the Defect Handling Process are covered in the Defect Triage Team section below.

2.1 Defect Triage Team

The scope for the Defect Triage Team is to disposition customer defects following the SLO guidelines, internal defects and all Beta defects reported on pre-GA releases. The Defect Triage core team for customer defects consists of the following representatives, and attendance is required:

- Quality Engineering Lead
- Product Manager
- Development Manager
- Customer Support & Training Representative (for customer defects).
- Product Security Champion (attendance mandatory for product security defects)

If a core team member is unable to attend the Defect Triage, then a backup representative with decision making authority should be appointed. The Quality Engineering Lead, Scrum Master or R&D Manager facilitates the Defect Triage meeting.

The following representatives are optional to attend the defect triage:

- Technical Writer is encouraged to attend the defect triage
- Professional Services Rep may attend at the request of the Technical Support Rep

- Usability Specialist for newly developed products, web products and/or significant UI changes
- Value Management Office (VMO) team members

The Defect Triage Team should refer to Defect Triage Process Guidelines and Best Practices for guidance to conduct the Defect Triage Meeting.

2.1.1 Development Manager

- a) The Development Manager is responsible for project schedule and associated resources and assigns defects to developers (moving defects to the Assigned or Approved state).

2.1.2 Quality Engineering Lead

- a) Responsible for dispositioning the defect based upon the decision of the Team:

2.1.3 The Developer

- a) Fixes the defect and integrates the code (note the 'change list' number in resolution tab)
- b) Updates the defect with applicable code change list identifier; indicate it is ready for code review.
- c) If applicable, attaches unit test results showing the defect is fixed and any applicable unit testing is completed.
- d) Evaluate documentation impact, as needed, based on the defect.

2.1.4 The Technical Writer

- a) Fixes/updates the assigned documentation defects.
- b) Updates the product documentation (if a defect fixed by Development has an impact on the documentation).

2.1.5 The Customer Support Representative

- a) Provides feedback from the Customer perspective. For some products/defects, the Customer Support representative should work with the Customer, Development & QE to help verify and close customer-specific defects.
- b) If a customer defect is not reproducible, then it is recommended to set up a video session with the customer to better understand the issue. Additionally, the supporting materials obtained from the customers should be provided to R&D to assist with the analysis and investigation.
- c) If there are any customer issues that should be escalated to the executive level, then these issues should be raised per the Executive Awareness/Escalation Procedure.

2.1.6 Product Manager

- a) Responsible for overall product "ownership". Responsible for making final decisions at Defect Triage meetings when there is disagreement between R&D

and Support.

- b) Chairs the Product Core Team, as needed, and has final authority on decisions to approve all enhancement requests, subject to any required executive approvals

2.1.7 The UX Lead

- a) Responsible for fixing and updating the design where the defect type is defined as UX.

3. Defect Processing Procedure

Defects can be reported either internally by the Product team or because of a customer request or Field Operations request through the Customer Support group. Refer to the QMS Glossary for the definitions of the applicable states for the defect process.

3.1 Defect Processing Workflow:

3.1.1 The reported defect is submitted and tracked to closure within the electronic defect management module of the Application Lifecycle Management system.

3.1.2 The submitted defect is assigned to a developer for investigation of the root cause of the issue and recommendation for disposition.

3.1.3 The submitted customer defect is triaged and dispositioned by the product team based on the investigation. The defect may be closed without any further actions (i.e., operator error, invalid defect, questions are answered) or approved to be fixed. All efforts should be made to have customer-generated defects go through the Triage process for general product releases. The defect should be approved to be fixed at the primary version of the current development release. Furthermore, if the defect fix is required for any supplemental releases, then the target version for fix record is submitted for each affected supplemental release version.

3.1.4 Resolving the approved defects – The assigned Developer, UX Lead or Tech Writer corrects the defect.

- For a Software defect, the fix undergoes development testing as needed to verify the correction. After development test, the Developer ensures that the modified files are checked-in prior to indicating that the defect has been implemented.
- UX defects are corrected by the UX Lead. Designs are updated and added to development for implementation.
- Documentation defects are corrected by the Technical Writer. Refer to section 3.6 for handling Documentation Defects.
- Once the fix is part of a build, the defect is pending verification test.

3.1.5 Verification of the implemented defect is performed by the Quality Engineer. If the

test to verify the defect fails, the defect is moved back for re-fix.

- If the code has been frozen due to expected release, the Development Manager and/or others may decide to move the defect fix into a future supplemental release.
- If needed, get the necessary customer feedback to close the defect.

3.1.6 The defect can be closed once the defect has been verified and all outstanding items have been resolved.

3.2 Defect Classifications

3.2.1 Defect Impact Classification

The Impact classification pertains specifically to the customer reported defects. This field is required only for customer reported defects. It is assigned based on the business impact to the customer. However, the defect Impact classification is not required to be assigned for all reported internal defects.

Impact Level	Impact Definition
1-Critical	Production System down. Unable to collect or generate critical data correctly. Critical project is at a stand-still.
2-Urgent	Cannot use major product feature or cannot fully utilize product as designed or documented. No "acceptable" workaround available. Necessary to correct major feature issue.
3-Important	General problem or question which does not prevent the user from fully utilizing product (as designed or documented).
4-Minor	Not awaiting immediate solution or response.

3.2.2 Defect Severity Classification

Defect Severity criticality is assessed for all internal and customer reported defects. The Severity classification captures the internal technical criticality assessment. Quality Engineering is responsible for classification the Severity of all incoming customer and internal defects.

Severity Level	Severity Definition
S1-Critical	Catastrophic defect that causes total failure of the software or unrecoverable data loss. Data Loss, Data Corruption, or Data Unavailable, crash/freeze of critical functionality Blocking the use of a critical functionality; blocking the execution of a critical workflow; main path does not work, workaround unavailable.
S2-Major	Defect results in severely impaired functionality or perceived as impaired by the user. Essential operations are disrupted but can still proceed. A workaround may exist, but its use is unsatisfactory in the long-term, or undiscoverable by the user.

	No Data Loss, data corruption, or loss of data availability. Features are not fully functional resulting in incorrect calculations or modeling results. Defined path of the use case doesn't fully work; alternate path does not work.
S3-Medium	Defect causes failure of non-critical aspects of the system. Essential operations are disrupted, but there is a reasonably satisfactory work around. No Data Loss, data corruption, or loss of data availability. Features are not fully functional resulting in inconsistent results. Non-critical issues; workaround available
S4-Minor	A non-essential operation is disrupted, but a work around exists. The impairment is slight and minor. Minor issues, Cosmetic issues
S5-Insignificant	No significant impact on operations

3.2.3 Defect Priority

There are many factors other than defect type that are considered when assigning a Priority. In the event where there is duplicate defects reported internally and externally by the customer, only one defect is required to be tracked to closure. In this case, the Priority of the defect must reflect the same priority level as the external defect.

Table 1: Priority Classification	
Priority	Priority Definition
1 - Showstopper	Production system down, unable to collect or generate critical data, critical project* at stand still for the customer.
2 - Critical	The customer is unable to use major product feature as designed or documented. Necessary to correct major feature issue.
3 - High	General problem or question, which does not prevent customer from fully utilizing product (as designed or documented)
4 - Medium	Customer is not awaiting immediate solution or response.

5 – Low	Enhancement. No significant impact on operations.
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3.3 Guidelines for Meeting SLO Response Times

The table defines the SLO response time goals for various types of defects. The R&D SLO measures the time from defect submission to a decision of whether the issue is a defect, and if so, whether it will be fixed.

Defect Priority	<u>Defect Definition</u> (Impact on Customer)	<u>Overall (Customer) SLO</u> (AT's commitment to the Customer, Target = 90%)	<u>R&D SLO</u> (R&D's commitment to CS&T, Target = 80%)
1 - Showstopper	Production system down, unable to collect or generate critical data, critical project at stand still	2 business days (response time within 1 hour)	Within 1 business day of defect submission
2 - Critical	Unable to use major product feature as designed or documented. Necessary to correct major feature issue	Within 1 week	Within 3 business days of defect submission
3 - High	General problem or question, which does not prevent customer from fuller utilizing product (as designed or documented)	Within 2 weeks	Within 5 business days of defect submission
4 – Medium	Not awaiting immediate solution or response	Within 5 weeks	Within 15 business days of defect submission

The R&D SLO timeline starts when a customer defect is submitted in the defect management system by CS&T. Showstopper (P1) and Critical (P2) defects must be evaluated immediately upon submission and cannot wait for the weekly Triage Review.

The R&D SLO timeline ends when the defect is changed to one of the following states.

- Assigned only when Work Process = "Waiting for Confirmation"
- Approved
- Implemented

- Pending Verification
- Verified
- Deferred
- Duplicate
- Closed

Note that the Assigned and Investigated states do not stop the SLO timeline, since a decision has not yet been determined.

If a Defect requires additional information from the customers, then the work process status should be set to indicate “Waiting for More Info”.

“Waiting for More Info” will be manually downgraded until the data/info is received from the customer

- P1/P2 is downgraded to P3 if no response is received within 3 business days
- P3 is downgraded to P4 if no response is received within 5 business days
- P4: If no response is received after 15 business days, then CS&T can recommend the defect to be closed

If a Defect requires additional investigation from R&D, then the work process status should be set to indicate “Investigating”.

For any opened defects, after 3 attempts spanning at least 30 calendar days where CS&T is requesting more information from the customer and if the customer is unresponsive, then the defect can be recommended by CS&T to be closed. CS&T can **open** a new defect when customer responds and still needs help.

3.4 Guidelines for Defect Root Cause Analysis

Root Cause analysis must be performed for all P1 and P2 defects reported by external customers. The defect triage team will evaluate the external P1/P2 defects and assign the applicable Root Cause to the respective customer defects. Below is an example of the defect Root Cause list. However, the formal list of Root Cause is maintained and controlled within the Azure DevOps Defect Tracking System.

Root Cause	Root Cause Sub List	Description
Coding	Error	Coding did not follow the design.
Design	Detail	The design lacked detail
Design	Missing	The design did not fully address the requirements
Design	Other	A design root cause not in this list.
Documentation	Missing	Documentation does not cover this feature
Documentation	Other	Documentation root cause not in this list.
Environment	Not supported	Product used in a non-supported environment
Install	Error	Install coding error
Install	Missing	Install does not cover this feature
Install	Other	An install root cause not in this list

Root Cause	Root Cause Sub List	Description
Internationalization/Localization	Layout	An issue in layout.
Internationalization/Localization	Linguistic	A linguistic issue.
Internationalization/Localization	OS or third-party apps	Issue with operation system or 3 rd party applications.
Internationalization/Localization	Other	
Internationalization/Localization	Regional Settings/Locales	Regional settings or locales issue with the localized product.
Internationalization/Localization	String Externalization	An issue with the string externalization.
Internationalization/Localization	String Handling	An issue with the string handling.
Maintenance	Algorithm	Limitations in existing algorithm
Maintenance	Other	Maintenance root cause not in this list
Maintenance	Resources	Limitations in resources assigned to maintenance
None		Root Cause Analysis has not been done. This is the default
Performance	Limitation	Product used in a manner that exceeds performance limitations
Requirement	Conflict with Existing Application	Requirement conflicts with existing application
Requirement	Detail	Requirement did not provide sufficient information
Requirement	Missing	No use case, architecture, or supplemental specification.
Requirement	Other	A requirement root cause not in this list.
Roadmap	Detail	Roadmap did not provide sufficient detail for use case, architecture and/or supplemental specifications
Roadmap	Not a feature	This is not a product feature.
Roadmap	Other	A roadmap root cause not in this list
Support	2nd Level	Issue with the 2 nd level support.
Test	Environment	Not tested in supported environment
Test	Integrated	Not tested in integrated environment
Test	Missing	Design not covered by Test Case
Test	Not Executed	Existing test case was not executed
Test	Other	A test root cause not in this list
Training	Customer	Customer requires additional training
Training	Other	A training root cause not in this list
Training	Development	R&D requires additional training
Training	Quality Engineering	QE requires additional training
Training	Support	Support requires additional training
Unit Test	Missing	Design not covered by unit testing
Unit Test	Not Executed	Existing unit test was not executed

Root Cause	Root Cause Sub List	Description
Unit Test	Other	A unit test root cause not in this list.

3.5 Guidelines for Defect Escalation

The triage team's decision on defect fixes needs to be consistent with AspenTech's official version support policies that have been published or have been incorporated into customer's software license and maintenance agreements. Customer Support will be responsible for providing accurate and complete information about relevant policies to the Triage Team for consideration in these decisions. For issues that have significant customer or field impact and needs to be escalated to the executive level, then these issues should be raised per the Executive Awareness/Escalation Procedure, P3705.

Additionally, when the customer defect is escalated to a L4 or L5 (refer to QMS Glossary), then a Corrective Action and Preventive Action (CAPA) should be initiated in the CAPA system to trigger a formal root cause analysis with relevant functional areas for the CAPA plan to remediate the escalated issue.

3.6 Documentation Defects

There are two different types of documentation defects 1) Documentation only issues and 2) Documentation impact issues/updates related to Software defects. For the documentation impact defects, the Developer and Technical Writer should collaborate on the documentation update(s) required. The Tech Writer completes the documentation updates when the software defect has been implemented. QE then verifies the documentation updates and software defect has been fixed to close out the defects.

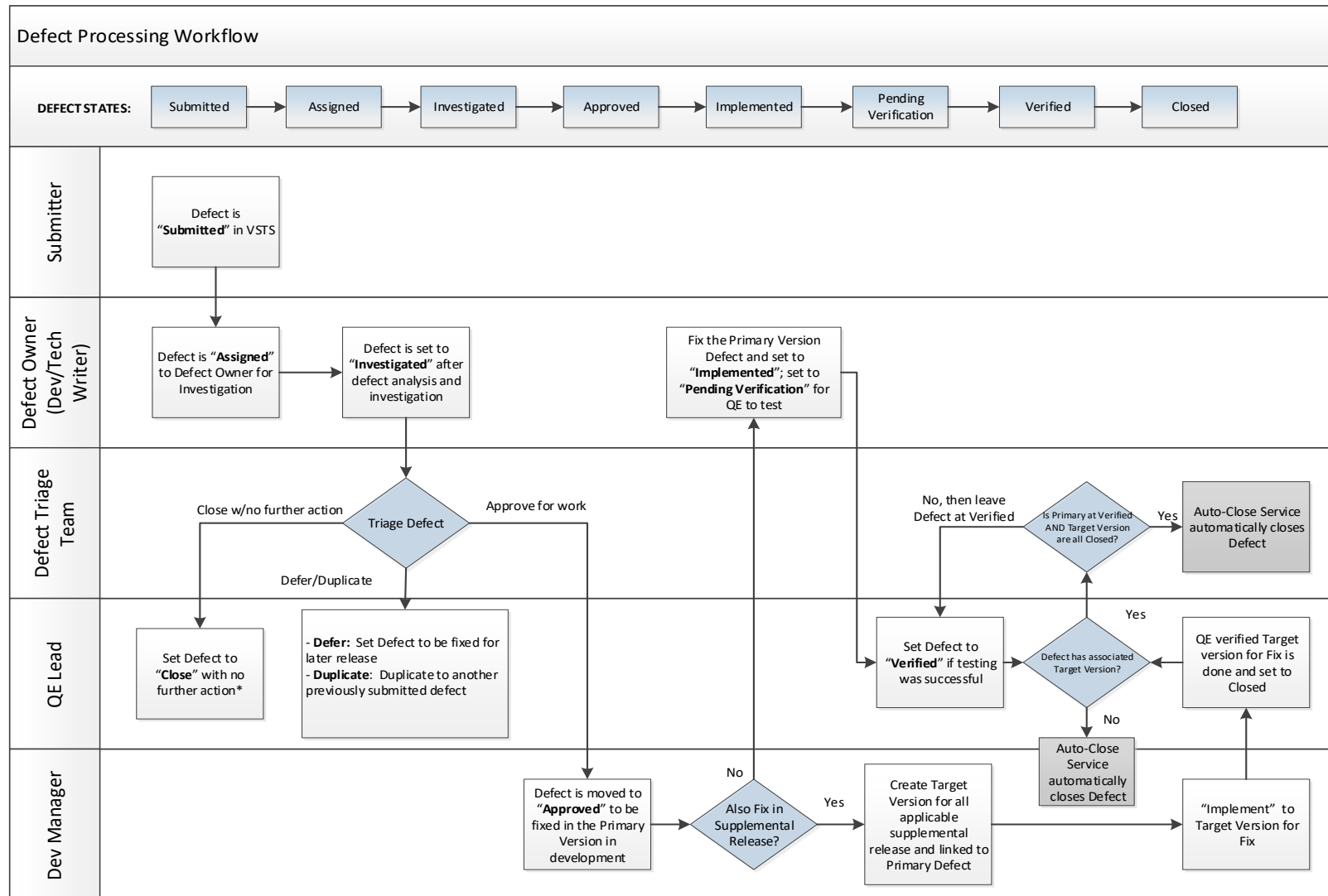
3.7 Emergency and Cumulative Patches

Showstopper and critical defects may be fixed between releases by issuing an Emergency or Cumulative Patch following the steps outlined in the appropriate procedure, P3702-Emergency Patch, and P3703-Cumulative Patch .

3.8 Handling of Deferred Defects from Previous Releases

At the beginning of each new release, the Product Manager is required to conduct a defect meeting to review any defect not addressed in a previous release. The Product Manager reviews the list of deferred defects and will decide the list of defects to be fixed into the RoadMap along with the necessary resource allocation.

4. Appendix - An example of the Defect Processing Lifecycle workflow



* Answered, No Plan to Fixed, Withdrawn, Not Reproducible, Workaround, Rejected